AI and Machine Learning Glossary

#	Term	Definition	Example sentence
1	**Artificial Intelligence (AI)**	The field of creating systems that can perform tasks requiring human-like intelligence.	*AI can analyze thousands of emails and classify them automatically.*
2	**Machine Learning (ML)**	A subset of AI where systems learn patterns from data rather than being explicitly programmed for every case.	*Machine learning can predict whether a transaction is fraudulent.*
3	**Deep Learning**	ML using neural networks with many layers to learn complex patterns.	*Deep learning models are widely used for image and speech recognition.*
4	**Neural Network**	A computational model inspired by the structure of the brain, consisting of interconnected nodes organized into layers.	*A neural network can learn to recognize objects in photographs.*
5	**Model**	A trained computational system that has learned patterns from data and can make predictions or generate outputs.	*We deployed a trained model to classify incoming documents.*
6	**Training**	The process of teaching a model by exposing it to data and adjusting its parameters.	*The model was trained using millions of examples.*
7	**Inference**	Using a trained model to produce a prediction, classification, or generated output.	*Inference takes two seconds for this particular model.*
8	**Dataset**	A collection of data used for training, validation, or testing a model.	*The team prepared a dataset containing 100,000 customer reviews.*

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9	**Feature**	An individual measurable property or input used by a machine-learning model.	*Age and income were two important features in the prediction model.*
10	**Label**	The expected answer or target associated with a training example.	*For spam detection, "spam" can be the label assigned to an email.*
11	**Embedding**	A numerical vector representation of data that captures semantic or meaningful relationships.	*We converted each document chunk into an embedding before storing it in the vector database.*
12	**Vector**	An ordered collection of numbers representing information in a mathematical space.	*The sentence was represented as a vector containing hundreds of dimensions.*
13	**Vector Database**	A database optimized for storing and searching vector representations.	*ChromaDB stores the document embeddings so we can retrieve relevant chunks.*
14	**Semantic Search**	Searching based on meaning rather than simply matching exact words.	*Semantic search can find "vehicle purchase" when the user searches for "buying a car."*
15	**LLM (Large Language Model)**	A large neural-network model trained on vast amounts of text to understand and generate language.	*An LLM can summarize a long report in a few paragraphs.*
16	**Token**	A unit of text processed by a language model; it may represent a word, part of a word, punctuation, or other text fragment.	*A model's context window limits how many tokens can be processed in one request.*

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17	**Prompt**	The instructions or input given to an AI model to produce a desired response.	*The developer improved the prompt to make the model return JSON consistently.*
18	**Context Window**	The maximum amount of information, measured in tokens, that a model can consider in a single interaction.	*The document was too large to fit into the model's context window.*
19	**RAG (Retrieval-Augmented Generation)**	A technique where relevant external information is retrieved and provided to an LLM before generating an answer.	*Our RAG system retrieves relevant policy documents before asking the LLM to answer the question.*
20	**Chunking**	Splitting a large document into smaller pieces for processing, embedding, and retrieval.	*We chunked the PDF into 800-word sections with overlap.*
21	**Fine-Tuning**	Further training a pre-trained model on specialized data to adapt it to a particular task or domain.	*The company fine-tuned the model using examples of its internal support conversations.*
22	**Hallucination**	When an AI model generates information that sounds plausible but is incorrect or unsupported.	*The RAG system was introduced to reduce hallucinations when answering questions about company policies.*
23	**Temperature**	A generation parameter that influences how predictable or varied a model's output is.	*A lower temperature is often useful when you want more consistent answers.*
24	**Classification**	Assigning an input to one or more predefined categories.	*The model classifies incoming emails as "General," "HR," or "Technical."*
25	**Cosine Similarity**	A mathematical measure used to determine how similar two vectors	*We used cosine similarity to find the document chunks*

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		are based on the angle between them.	*most relevant to the user's question*.